CMSC216: Assembly Basics and x86-64

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Logistics

Reading Bryant/O'Hallaron

- Ch 3.1-7: Assembly, Arithmetic, Control
- Later Ch 3.8-11: Arrays, Structs, Floats
- Any overview guide to x86-64 assembly instructions such as Brown University's x64 Cheat Sheet

Assignments

- Project 2 Fri (Bits, GDB)
- Lab05 Sun (Asm Basics)

Goals

- x86-64 Overview
- Assembly Arithmetic

*Your future self will thank you for writing **simple** C code for your P2 Thermometer implementation!*

Announcements

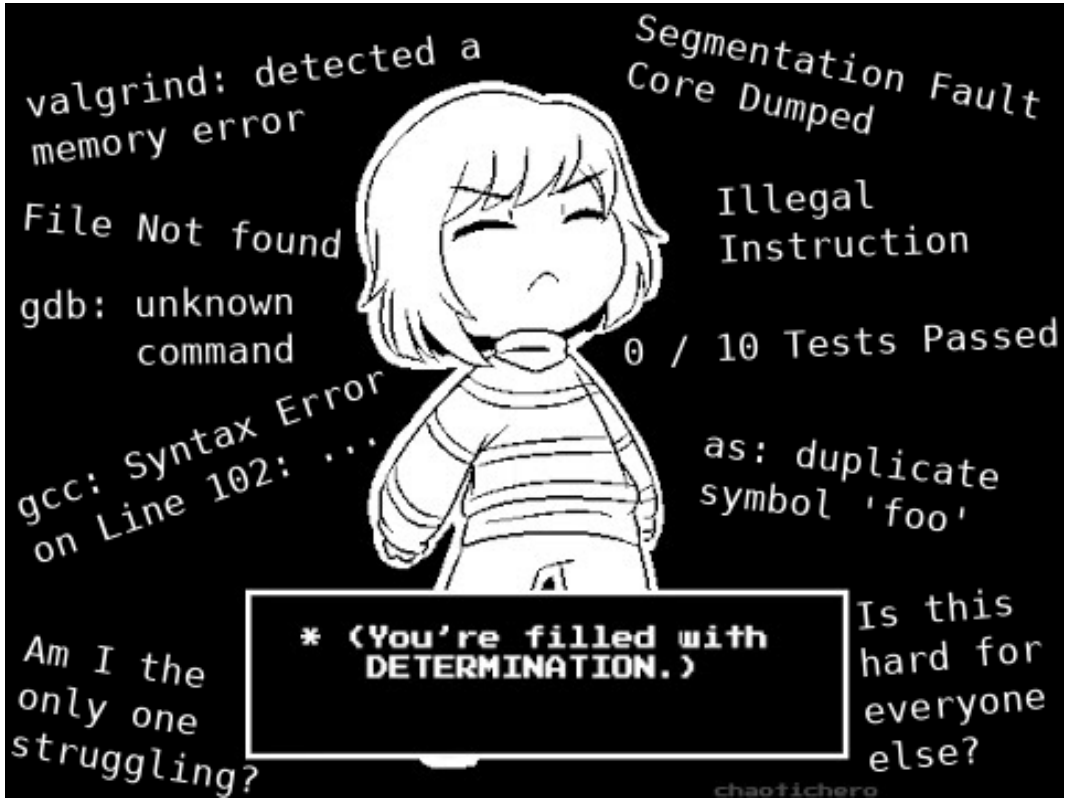
Technica Recruiting Organizers

See Piazza Post 115 and consider applying

Optional Floating Point Video Lecture

Posted and linked from the schedule, watch if moved to do so,
Read Bryant/O'Hallaron Ch 2.4 for more details

Don't Give Up, Stay Determined!



- If Project 1 / Exam 1 went awesome, count yourself lucky
- If things did not go well, Don't Give Up
- Spend some time contemplating why things didn't go well, talk to course staff about it, learn from mistakes
- There is a LOT of semester left and time to recover from a bad start

Wrapping Up Integer Affairs

Pre-exam one slides contain info on byte-ordering (little / big endian) and bit-wise operations to be discussed here prior to assembly.

The **Many** Assembly Languages

- Most **microprocessors** are created to understand a **binary machine language**
- Machine Language provides means to manipulate internal memory, perform arithmetic, etc.
- The Machine Language of one processor is **not understood** by other processors

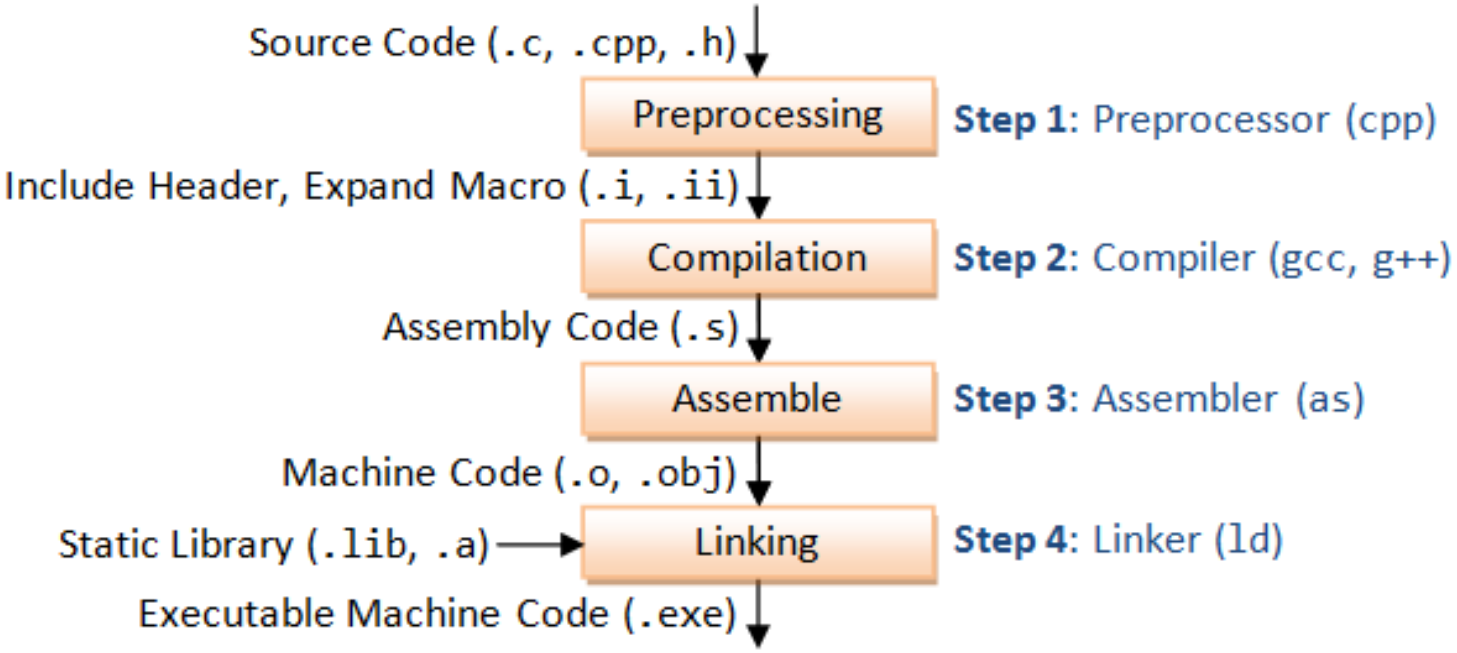
MOS Technology 6502

- 8-bit operations, limited addressable memory, **1 general purpose register**, powered notable gaming systems in the 1980s
- Apple IIe, Atari 2600, Commodore
- Nintendo Entertainment System / Famicom

IBM Cell Microprocessor

- Developed in early 2000s, 64-bit, many cores (execution elements), many registers (32 on the PPE), large addressable space, fast multimedia performance, is a **pain** to program
- Playstation 3 and Blue Gene Supercomputer

Assemblers and Compilers



- **Compiler**: chain of tools that translate high level languages to lower ones, may perform optimizations
- **Assembler**: translates text description of the machine code to binary, formats for execution by processor, late compiler stage
- Consequence: The compiler can **generate assembly code**
- Generated assembly is a pain to read but is often quite fast
- Consequence: A compiler on an Intel chip can generate assembly code for a different processor, **cross compiling**

Our focus: The x86-64 Assembly Language

- x86-64 Targets Intel/AMD chips with 64-bit word size
  Reminder: 64-bit "word size" ≈ size of pointers/addresses
- Lineage of x86 family
  - 1970s: 16-bit systems like Intel 8086
  - 1990s: IA32 (Intel 32-bit systems like 80386 and 80486)
  - 2000s: x86-64 (64-bit extension by AMD)
- x86-64 is backwards compatibility, consequently much cruft
  - Can run compiled code from the 70's / 80's on modern processors without much trouble BUT means 50-year-old instructions must be preserved
  - x86-64 is not the assembly language you would design from scratch today, it's the assembly you have to code against
  - RISC-V is a new assembly language that is "clean" as it has no history to support (and few CPUs run it)
- Warning: Lots of information available on the web for Intel assembly programming **BUT** some of it is dated, IA32 info which may not work on 64-bit systems

x86-64 Assembly Language Syntax(es)

- Different assemblers understand different syntaxes for the same assembly language
- GCC use the GNU Assembler (GAS, command `'as file.s'`)
- GAS and Textbook favor AT&T syntax so **we will too**
- NASM assembler favors Intel, may see this online

AT&T Syntax (Our Focus)

```
multstore:
        pushq   %rbx
        movq    %rdx, %rbx
        call    mult2@PLT
        movq    %rax, (%rbx)
        popq    %rbx
        ret
```

- Use of `%` to indicate registers
- Use of `q/l/w/b` to indicate 64 / 32 / 16 / 8-bit operands

Intel Syntax

```
multstore:
        push    rbx
        mov     rbx, rdx
        call    mult2@PLT
        mov     QWORD PTR [rbx], rax
        pop     rbx
        ret
```

- Register names are bare
- Use of `QWORD` etc. to indicate operand size

Generating Assembly from C Code

- `gcc -S file.c` will stop compilation at assembly generation
- Leaves assembly code in `file.s`
  - `file.s` and `file.S` conventionally assembly code though sometimes `file.asm` is used
- By default, compiler generates code that is often difficult for humans to interpret, may include re-arrangements, "conservative" compatibility assembly, etc. increasing size of assembly considerably
- `gcc -Og file.c`: optimize for debugging, generally makes it easier to read generated assembly, aligns somewhat more closely to C code

Example of Generating Assembly from C

```
>> cat exchange.c                          # show C file to be translated
// exchange.c: sample C function
// to compile to assembly
long exchange(long *xp, long y){           # function to translate
  long x = *xp;                            # involves pointer deref
  *xp = y;
  return x;
}

>> gcc -Og -S exchange.c                   # Compile to show assembly
                                           # -Og: debugging level optimization
                                           # -S: only output assembly

>> cat exchange.s                          # show assembly output
        .file   "exchange.c"
        .text
        .globl  exchange
        .type   exchange, @function
exchange:                                  # beginning of exchange function
.LFB0:
        .cfi_startproc
        movq    (%rdi), %rax               # pointer derefs in assembly
        movq    %rsi, (%rdi)               # uses registers
        ret
        .cfi_endproc
.LFE0:
        .size   exchange, .-exchange
        .ident  "GCC: (GNU) 11.1.0"
        .section        .note.GNU-stack,"",@progbits
```

gcc -Og -S mstore.c

```
> cat mstore.c                                  # show a C file
long mult2(long a, long b);
void multstore(long x, long y, long *dest){
  long t = mult2(x, y);
  *dest = t;
}

> gcc -Og -S mstore.c                           # Compile to show assembly
                                                # -Og: debugging level optimization
                                                # -S: only output assembly

> cat mstore.s                                  # show assembly output
        .file   "mstore.c"
        .text
        .globl  multstore                       # function symbol for linking
        .type   multstore, @function
multstore:                                      # beginning of mulstore function
.LFB0:
        .cfi_startproc                          # assembler directives
        pushq   %rbx                            # assembly instruction
        .cfi_def_cfa_offset 16                  # directives
        .cfi_offset 3, -16
        movq    %rdx, %rbx                      # assembly instructions
        call    mult2@PLT                       # function call
        movq    %rax, (%rbx)
        popq    %rbx
        .cfi_def_cfa_offset 8
        ret                                     # function return
        .cfi_endproc
```

Every Programming Language

Look for the following as it should almost always be there

- ☐ Comments
- ☐ Statements/Expressions
- ☐ Variable Types
- ☐ Assignment
- ☐ Basic Input/Output
- ☐ Function Declarations
- ☐ Conditionals (if-else)
- ☐ Iteration (loops)
- ☐ Aggregate data (arrays, structs, objects, etc)
- ☐ Library System

Exercise: Examine `col_simple_asm.s`

Take a simple sample problem to demonstrate assembly:

> *Computes Collatz Sequence starting at n=10:*
> *if n is ODD n=n*3+1; else n=n/2.*
> *Return the number of steps to converge to 1 as the* ***return code*** *from* `main()`

The following codes solve this problem

Code	Notes
`col_simple_asm.s`	Hand-coded assembly for obvious algorithm
	Straight-forward reading
`col_unsigned.c`	Unsigned C version
	Generated assembly is reasonably readable
`col_signed.c`	Signed C vesion
	Generated assembly is ... interesting

- Kauffman will Compile/Run code
- Students should study the code and predict what lines do
- Illustrate tricks associated with gdb and assembly

Exercise: col_simple_asm.s

```
### Compute Collatz sequence starting at 10 in assembly.
.text
.global main
main:
        movl    $0,  %r8d       # int steps = 0;
        movl    $10, %ecx       # int n = 10;
.LOOP:
        cmpl    $1,  %ecx       # while(n > 1){ // immediate must be first
        jle     .END            #   n <= 1 exit loop
        movl    $2,  %esi       #   divisor in esi
        movl    %ecx,%eax       #   prep for division: must use edx:eax
        cltd                    #   division prep: extend sign from eax to edx
        idivl   %esi            #   divide edx:eax by esi
                                #   eax has quotient, edx remainder
        cmpl    $1,%edx         #   if(n % 2 == 1) {
        jne     .EVEN           #     not equal, go to even case
.ODD:
        imull   $3, %ecx        #     n = n * 3
        incl    %ecx            #     n = n + 1 OR n++
        jmp     .UPDATE         #   }
.EVEN:                          #   else{
        sarl    $1,%ecx         #     n = n / 2; via right shift
.UPDATE:                        #   }
        incl    %r8d            #   steps++;
        jmp     .LOOP           # }
.END:
        movl    %r8d, %eax      # r8d is steps, move to eax for return value
        ret
```

Answers: x86-64 Assembly Basics for AT&T Syntax

- *Comments* are one-liners starting with #
- *Statements*: each line does ONE thing, frequently text representation of an assembly instruction

```
movq    %rdx, %rbx      # move rdx register to rbx
```

- Assembler directives and labels are also possible:

```
.global  multstore        # notify linker of location multstore
multstore:                # label beginning of multstore section
      blah blah blah      # instructions in this this section
```

- *Variables*: mainly **registers**, also memory ref'd by registers maybe some named global locations
- *Assignment*: instructions like `movX` that put bits into registers and memory
- *Conditionals/Iteration*: assembly instructions that jump to code locations
- *Functions*: code locations that are **labeled** and global
- *Aggregate data*: none, use the stack/multiple registers
- *Library System*: link to other code

So what *are* these Registers?

- Memory locations directly wired to the CPU
- Usually *very* fast to access, faster than **main memory**
- Most instructions involve registers, access or change reg val

Example: Adding Together Integers

- Ensure registers have desired values in them
- Issue an `addX` instruction involving the two registers
- Result will be stored in a register

```
addl %eax, %ebx
# add ints in eax and ebx,  store result in ebx

addq %rcx, %rdx
# add longs in rcx and rdx, store result in rdx
```

- Note instruction and register names indicate whether 32-bit `int` or 64-bit `long` are being added

x86-64 "General Purpose" Registers

Many "general purpose" registers have special purposes and conventions associated such as

- Return Value: `%rax` / `%eax` / `%ax`
- Function Args 1 to 6: `%rdi`, `%rsi`, `%rdx`, `%rcx`, `%r8`, `%r9`
- Stack Pointer (top of stack): `%rsp`
- Old Code Base Pointer: `%rbp`, historically start of current stack frame but is not used that way in modern codes

Note: There are also Special Registers like `%rip` and `%eflags` which we will discuss later.

64-bit	32-bit	16-bit	8-bit	Notes
`%rax`	`%eax`	`%ax`	`%al`	Return Val
`%rbx`	`%ebx`	`%bx`	`%bl`	
`%rcx`	`%ecx`	`%cx`	`%cl`	Arg 4
`%rdx`	`%edx`	`%dx`	`%dl`	Arg 3
`%rsi`	`%esi`	`%si`	`%sil`	Arg 2
`%rdi`	`%edi`	`%di`	`%dil`	Arg 1
`%rsp`	`%esp`	`%sp`	`%spl`	Stack Ptr
`%rbp`	`%ebp`	`%bp`	`%bpl`	Base Ptr?
`%r8`	`%r8d`	`%r8w`	`%r8b`	Arg 5
`%r9`	`%r9d`	`%r9w`	`%r9b`	Arg 6
`%r10`	`%r10d`	`%r10w`	`%r10b`	
`%r11`	`%r11d`	`%r11w`	`%r11b`	
`%r12`	`%r12d`	`%r12w`	`%r12b`	
`%r13`	`%r13d`	`%r13w`	`%r13b`	
`%r14`	`%r14d`	`%r14w`	`%r14b`	
`%r15`	`%r15d`	`%r15w`	`%r15b`	

Caller Save:	Restore after calling func
Callee Save:	Restore before returning

Register Naming Conventions

- AT&T syntax identifies registers with prefix %
- Naming convention is a historical artifact
- Originally 16-bit architectures in x86 had
  - General registers `ax,bx,cx,dx`,
  - Special Registers `si,di,sp,bp`
- *Extended* to 32-bit: `eax,ebx,...,esi,edi,...`
- Grew again to 64-bit: `rax,rbx,...,rsi,rdi,...`
- Added Eight 64-bit regs `r8,r9,...,r14,r15` with 32-bit portion `r8d,r9d,...`, 16-bit `r8w,r9w...`, etc.
- Instructions must match registers sizes:

```
addw %ax,  %bx  #      word (16-bit)
addl %eax, %ebx # long word (32-bit)
addq %rax, %rbx # quad-word (64-bit)
```

- When hand-coding assembly, easy to mess this up, assembler will error out

Hello World in x86-64 Assembly : Not that Easy

- Non-trivial in assembly because **output is involved**
  - Try writing `helloworld.c` without `printf()`
- Output is the business of the **operating system**, always a request to the almighty OS to put something somewhere
  - **Library call**: `printf("hello");` mangles some bits but eventually results with a ...
  - **System call**: Unix system call directly implemented in the OS **kernel**, puts bytes into files / onto screen as in

    ```
    write(1, buf, 5);  // file 1 is screen output
    ```

This gives us several options for hello world in assembly:

1. `hello_printf64.s`: via calling `printf()` which means the C standard library must be (painfully) linked
2. `hello64.s` via direct system `write()` call which means no external libraries are needed: OS knows how to write to files/screen. Use the 64-bit Linux calling convention.
3. `hello32.s` via direct system call using the older 32 bit Linux calling convention which "traps" to the operating system.

(Optional): The OS Privilege: System Calls

- Most interactions with the outside world happen via Operating System Calls (or just "system calls")
- User programs indicate what service they want performed by the OS via making system calls
- System Calls differ for each language/OS combination
  - x86-64 Linux: set `%rax` to system call number, set other args in registers, issue `syscall`
  - IA32 Linux: set `%eax` to system call number, set other args in registers, issue an **interrupt**
  - C Code on Unix: make system calls via `write()`, `read()` and others (studied in CSCI 4061)
  - Tables of Linux System Call Numbers
    - 64-bit (335 calls)
    - 32-bit (190 calls)
  - Mac OS X: very similar to the above (it's a Unix)
  - Windows: use OS wrapper functions
- OS executes **priveleged** code that can manipulate any part of memory, touch internal data structures corresponding to files, do other fun stuff discussed in OS courses

Basic Instruction Classes

- **Remember:** Goal is to understand assembly as a *target* for higher languages, not become expert "assemblists"
- Means we won't hit all 4,834 pages of the Intel x86-64 Manual
- Brown University's x64 Cheat Sheet has a good overview
- x86 Assembly Guide from Yale is also good but is limited to 32-bit coverage

Kind	Assembly Instructions
Fundamentals	
- Memory Movement	`mov`
- Stack manipulation	`push,pop`
- Addressing modes	`(%eax),12(%eax,%ebx)...`
Arithmetic/Logic	
- Arithmetic	`add,sub,mul,div,lea`
- Bitwise Logical	`and,or,xor,not`
- Bitwise Shifts	`sal,sar,shr`
Control Flow	
- Compare / Test	`cmp,test`
- Set on result	`set`
- Jumps (Un)Conditional	`jmp,je,jne,jl,jg,...`
- Conditional Movement	`cmove,cmovg,...`
Procedure Calls	
- Stack manipulation	`push,pop`
- Call/Return	`call,ret`
- System Calls	`syscall`
Floating Point Ops	
- FP Reg Movement	`vmov`
- Conversions	`vcvts`
- Arithmetic	`vadd,vsub,vmul,vdiv`
- Extras	`vmins,vmaxs,sqrts`

Data Movement: movX instruction

```
movX SOURCE, DEST    # move/copy source value to dest
```

Overview

- Moves data...
  - Reg to Reg
  - Mem to Reg
  - Reg to Mem
  - Imm to ...
- Reg: register
- Mem: main memory
- Imm: "immediate" value (constant) specified like
  - `$21` : decimal
  - `$0x2f9a` : hexadecimal
  - **NOT** `1234` (mem adder)
- More info on operands next

Examples

```
## 64-bit quadword moves
movq $4,  %rbx     # rbx = 4;
movq %rbx,%rax     # rax = rbx;
movq $10, (%rcx)   # *rcx = 10;

## 32-bit longword moves
movl $4,  %ebx     # ebx = 4;
movl %ebx,%eax     # eax = ebx;
movl $10, (%rcx)   # *rcx = 10;
```

Note variations

- `movq` for 64-bit (8-byte)
- `movl` for 32-bit (4-byte)
- `movw` for 16-bit (2-byte)
- `movb` for 8-bit (1-byte)

Operands and Addressing Modes

In many instructions like `movX`, operands can have a variety of forms called **addressing modes**, may include constants and memory addresses

Style	Address Mode	C-like	Notes
`$21` `$0xD2`	immediate	`21`	value of constant like 21 or `0xD2 = 210`
`%rax`	register	`rax`	to/from register contents
`(%rax)`	indirect	`*rax`	reg holds memory address, deref
`8(%rax)` `4(%rdx)`	displaced	`*(rax+2)` `rdx->field`	base plus constant offset, often used for strcut field derefs
`(%rax,%rbx)`	indexed	`*(rax+rbx)` `char_arr[rbx]`	base plus offset in given reg actual value of `rbx` is used, NOT multiplied by `sizeof()`
`(%rax,%rbx,4)` `(%rax,%rbx,8)`	scaled index	`rax[rbx]` `rax[rbx]`	like array access with `sizeof(..)=4` “” with `sizeof(..)=8`
`1024`	absolute	...	Absolute address #1024 **Rarely used**

Exercise: Show movX Instruction Execution

Code movX_exercise.s

```
movl $16, %eax
movl $20, %ebx
movq $24, %rbx
## POS A

movl %eax,%ebx
movq %rcx,%rax
## POS B

movq $45,(%rdx)
movl $55,16(%rdx)
## POS C

movq $65,(%rcx,%rbx)
movq $3,%rbx
movq $75,(%rcx,%rbx,8)
## POS D
```

Registers/Memory

```
INITIAL
|-----+-------+-------|
| REG | %rax  |     0 |
|     | %rbx  |     0 |
|     | %rcx  | #1024 |
|     | %rdx  | #1032 |
|-----+-------+-------|
| MEM | #1024 |    35 |
|     | #1032 |    25 |
|     | #1040 |    15 |
|     | #1048 |     5 |
|-----+-------+-------|
```

Lookup...

May need to look up addressing conventions for things like...

```
movX %y,%x     # reg y to reg x
movX $5,(%x)   # 5 to address in %x
```

Answers Part 1/2: `movX` Instruction Execution

```
                    movl $16, %eax
                    movl $20, %ebx       movl %eax,%ebx
                    movq $24, %rbx       movq %rcx,%rax #WARNING!
INITIAL             ## POS A             ## POS B
|-------+-------|   |-------+-------|   |-------+-------|
| REG   | VALUE |   | REG   | VALUE |   | REG   | VALUE |
| %rax  |     0 |   | %rax  |    16 |   | %rax  | #1024 |
| %rbx  |     0 |   | %rbx  |    24 |   | %rbx  |    16 |
| %rcx  | #1024 |   | %rcx  | #1024 |   | %rcx  | #1024 |
| %rdx  | #1032 |   | %rdx  | #1032 |   | %rdx  | #1032 |
|-------+-------|   |-------+-------|   |-------+-------|
| MEM   | VALUE |   | MEM   | VALUE |   | MEM   | VALUE |
| #1024 |    35 |   | #1024 |    35 |   | #1024 |    35 |
| #1032 |    25 |   | #1032 |    25 |   | #1032 |    25 |
| #1040 |    15 |   | #1040 |    15 |   | #1040 |    15 |
| #1048 |     5 |   | #1048 |     5 |   | #1048 |     5 |
|-------+-------|   |-------+-------|   |-------+-------|
```

`#WARNING!`: On 64-bit systems, ALWAYS use a 64-bit reg name like `%rdx` and `movq` to copy memory addresses; using smaller name like `%edx` will miss half the memory addressing leading to major memory problems

Answers Part 2/2: `movX` Instruction Execution

```
movl %eax,%ebx
movq %rcx,%rax #!
```

```
 ## POS B
|-------+-------|
| REG   | VALUE |
| %rax  | #1024 |
| %rbx  |    16 |
| %rcx  | #1024 |
| %rdx  | #1032 |
|-------+-------|
| MEM   | VALUE |
| #1024 |    35 |
| #1032 |    25 |
| #1040 |    15 |
| #1048 |     5 |
|-------+-------|
```

```
movq $45,(%rdx)
          #1032
movq $55,16(%rdx)
         16+#1032=#1048
```

```
## POS C
|-------+-------|
| REG   | VALUE |
| %rax  | #1024 |
| %rbx  |    16 |
| %rcx  | #1024 |
| %rdx  | #1032 |
|-------+-------|
| MEM   | VALUE |
| #1024 |    35 |
| #1032 |    45 |
| #1040 |    15 |
| #1048 |    55 |
|-------+-------|
```

```
movq $65,(%rcx,%rbx)
          #1024+16 = #1040
movq $3,%rbx
movq $75,(%rcx,%rbx,8)
          #1024 + 3*8 = #1048
```

```
## POS D
|-------+-------|
| REG   | VALUE |
| %rax  | #1024 |
| %rbx  |     3 |
| %rcx  | #1024 |
| %rdx  | #1032 |
|-------+-------|
| MEM   | VALUE |
| #1024 |    35 |
| #1032 |    45 |
| #1040 |    65 |
| #1048 |    75 |
|-------+-------|
```

gdb Assembly: Examining Memory

gdb commands `print` and `x` allow one to print/examine memory memory of interest. Try on `movX_exercises.s`

```
(gdb) tui enable              # TUI mode
(gdb) layout asm              # assembly mode
(gdb) layout reg              # show registers
(gdb) stepi                   # step forward by single Instruction
(gdb) print $rax              # print register rax
(gdb) print *($rdx)           # print memory pointed to by rdx
(gdb) print (char *) $rdx     # print as a string (null terminated)
(gdb) x $r8                   # examine memory at address in r8
(gdb) x/3d $r8                # same but print as 3 4-byte decimals
(gdb) x/6g $r8                # same but print as 6 8-byte decimals
(gdb) x/s  $r8                # print as a string (null terminated)
(gdb) print *((int*) $rsp)    # print top int on stack (4 bytes)
(gdb) x/4d $rsp               # print top 4 stack vars as ints
(gdb) x/4x $rsp               # print top 4 stack vars as ints in hex
```

Many of these tricks are needed to debug assembly.

Register Size and Data Movement

- %rax is 64-bit register, %eax is its lower 32 bits
- Data movement involving small registers **may NOT overwrite** higher bits in extended register
- Moving data to low 32-bit regs automatically zeros high 32-bits

```
movabsq $0x1122334455667788, %rax  # 8 bytes to %rax
movl $0xAABBCCDD, %eax             # 4 bytes to %eax
## %rax is now 0x00000000AABBCCDD
```

- Moving data to other small regs DOES NOT ALTER high bits

```
movabsq $0x1122334455667788, %rax  # 8 bytes to %rax
movw $0xAABB, %ax                  # 2 bytes to %ax
## %rax is now 0x112233445566AABB
```

- Gives rise to two other families of movement instructions for moving little registers (X) to big (Y) registers, see movz_examples.s

```
## movzXY move zero extend, movsXY move sign extend
movabsq $0x112233445566AABB,%rdx
movzwq %dx,%rax       # %rax is 0x000000000000AABB
movswq %dx,%rax       # %rax is 0xFFFFFFFFFFFFAABB
```

Exercise: movX differences in Main Memory

Instr	# bytes
`movb`	1 byte
`movw`	2 bytes
`movl`	4 bytes
`movq`	8 bytes

Show the result of each of the following copies to main memory in sequence.

```
movl   %eax,   (%rsi) #1
movq   %rax,   (%rsi) #2
movb   %cl,    (%rsi) #3
movw   %cx,   2(%rsi) #4
movl   %ecx, 4(%rsi) #5
movw   4(%rsi), %ax   #6
```

```
INITIAL
|-------+--------------------|
| REG   |                    |
| rax   | 0x00000000DDCCBBAA |
| rcx   | 0x000000000000FFEE |
| rsi   |              #1024 |
|-------+--------------------|
| MEM   |                    |
| #1024 |               0x00 |
| #1025 |               0x11 |
| #1026 |               0x22 |
| #1027 |               0x33 |
| #1028 |               0x44 |
| #1029 |               0x55 |
| #1030 |               0x66 |
| #1031 |               0x77 |
| #1032 |               0x88 |
| #1033 |               0x99 |
|-------+--------------------|
```

Answers: movX to Main Memory 1/2

```
|-----+--------------------|  movl   %eax,  (%rsi) #1 4 bytes rax -> #1024
| REG |                    |  movq   %rax,  (%rsi) #2 8 bytes rax -> #1024
| rax | 0x00000000DDCCBBAA |  movb   %cl,   (%rsi) #3 1 byte  rcx -> #1024
| rcx | 0x000000000000FFEE |  movw   %cx,  2(%rsi) #4 2 bytes rcx -> #1026
| rsi |              #1024 |  movl   %ecx, 4(%rsi) #5 4 bytes rcx -> #1028
|-----+--------------------|  movw   4(%rsi), %ax  #6 2 bytes #1024 -> rax
```

```
                    #1                    #2                    #3
INITIAL             movl %eax,(%rsi)      movq %rax,(%rsi)      movb %cl,(%rsi)
|-------+------|    |-------+------|      |-------+------|      |-------+------|
| MEM   |      |    | MEM   |      |      | MEM   |      |      | MEM   |      |
| #1024 | 0x00 |    | #1024 | 0xAA |      | #1024 | 0xAA |      | #1024 | 0xEE |
| #1025 | 0x11 |    | #1025 | 0xBB |      | #1025 | 0xBB |      | #1025 | 0xBB |
| #1026 | 0x22 |    | #1026 | 0xCC |      | #1026 | 0xCC |      | #1026 | 0xCC |
| #1027 | 0x33 |    | #1027 | 0xDD |      | #1027 | 0xDD |      | #1027 | 0xDD |
| #1028 | 0x44 |    | #1028 | 0x44 |      | #1028 | 0x00 |      | #1028 | 0x00 |
| #1029 | 0x55 |    | #1029 | 0x55 |      | #1029 | 0x00 |      | #1029 | 0x00 |
| #1030 | 0x66 |    | #1030 | 0x66 |      | #1030 | 0x00 |      | #1030 | 0x00 |
| #1031 | 0x77 |    | #1031 | 0x77 |      | #1031 | 0x00 |      | #1031 | 0x00 |
| #1032 | 0x88 |    | #1032 | 0x88 |      | #1032 | 0x88 |      | #1032 | 0x88 |
| #1033 | 0x99 |    | #1033 | 0x99 |      | #1033 | 0x99 |      | #1033 | 0x99 |
|-------+------|    |-------+------|      |-------+------|      |-------+------|
```

Answers: movX to Main Memory 2/2

```
|-----+--------------------|  movl   %eax,  (%rsi) #1 4 bytes rax -> #1024
| REG |                    |  movq   %rax,  (%rsi) #2 8 bytes rax -> #1024
| rax | 0x00000000DDCCBBAA |  movb   %cl,   (%rsi) #3 1 byte  rcx -> #1024
| rcx | 0x000000000000FFEE |  movw   %cx,  2(%rsi) #4 2 bytes rcx -> #1026
| rsi |              #1024 |  movl   %ecx, 4(%rsi) #5 4 bytes rcx -> #1028
|-----+--------------------|  movw   4(%rsi), %ax  #6 2 bytes #1024 -> rax
```

```
#3                  #4                  #5                  #6
movb %cl,(%rsi)     movw %cx,2(%rsi)    movl %ecx,4(%rsi)   movw 4(%rsx),%ax
|-------+------|    |-------+------|    |-------+------|    |-------+------|
| MEM   |      |    | MEM   |      |    | MEM   |      |    | MEM   |      |
| #1024 | 0xEE |    | #1024 | 0xEE |    | #1024 | 0xEE |    | #1024 | 0xEE |
| #1025 | 0xBB |    | #1025 | 0xBB |    | #1025 | 0xBB |    | #1025 | 0xBB |
| #1026 | 0xCC |    | #1026 | 0xEE |    | #1026 | 0xEE |    | #1026 | 0xEE |
| #1027 | 0xDD |    | #1027 | 0xFF |    | #1027 | 0xFF |    | #1027 | 0xFF |
| #1028 | 0x00 |    | #1028 | 0x00 |    | #1028 | 0xEE |    | #1028 | 0xEE |<
| #1029 | 0x00 |    | #1029 | 0x00 |    | #1029 | 0xFF |    | #1029 | 0xFF |<
| #1030 | 0x00 |    | #1030 | 0x00 |    | #1030 | 0x00 |    | #1030 | 0x00 |
| #1031 | 0x00 |    | #1031 | 0x00 |    | #1031 | 0x00 |    | #1031 | 0x00 |
| #1032 | 0x88 |    | #1032 | 0x88 |    | #1032 | 0x88 |    | #1032 | 0x88 |
| #1033 | 0x99 |    | #1033 | 0x99 |    | #1033 | 0x99 |    | #1033 | 0x99 |
|-------+------|    |-------+------|    |-------+------|    |-------+------|
```

```
| rax | 0x00000000DDCCFFEE |
```

addX : A Quintessential ALU Instruction

```
addX  B, A     # A = A+B

OPERANDS:
addX %reg,  %reg
addX (%mem),%reg
addX %reg,  (%mem)
addX $con,  %reg
addX $con,  (%mem)

# No mem+mem or con+con
```

- Addition represents most 2-operand ALU instructions well
- Second operand `A` is modified by first operand `B`, No change to `B`
- Variety of register, memory, constant combinations honored
- `addX` has variants for each register size: `addq, addl, addw, addb`

```
EXAMPLES:
addq %rdx,  %rcx       # rcx = rcx + rdx
addl %eax,  %ebx       # ebx = ebx + eax
addq $42,   %rdx       # rdx = rdx + 42
addl (%rsi),%edi       # edi = edi + *rsi
addw %ax,   (%rbx)     # *rbx = *rbx + ax
addq $55,   (%rbx)     # *rbx = *rbx + 55

addl (%rsi,%rax,4),%edi    # edi = edi+rsi[rax] (int)
```

Optional Exercise: Addition

Show the results of the following addX/movX ops at each of the specified positions

```
addq $1,%rcx        # con + reg
addq %rbx,%rax      # reg + reg
## POS A

addq (%rdx),%rcx    # mem + reg
addq %rbx,(%rdx)    # reg + mem
addq $3,(%rdx)      # con + mem
## POS B

addl $1,(%r8,%r9,4)      # con + mem
addl $1,%r9d             # con + reg
addl %eax,(%r8,%r9,4)    # reg + mem
addl $1,%r9d             # con + reg
addl (%r8,%r9,4),%eax    # mem + reg
## POS C
```

INITIAL

REGS	
%rax	15
%rbx	20
%rcx	25
%rdx	#1024
%r8	#2048
%r9	0
MEM	
#1024	100
...	...
#2048	200
#2052	300
#2056	400

Answers: Addition

```
INITIAL            POS A              POS B              POS C
|-------+-------|  |-------+-------|  |-------+-------|  |-------+-------|
| REG   |       |  | REG   |       |  | REG   |       |  | REG   |       |
| %rax  |    15 |  | %rax  |    35 |  | %rax  |    35 |  | %rax  |   435 |
| %rbx  |    20 |  | %rbx  |    20 |  | %rbx  |    20 |  | %rbx  |    20 |
| %rcx  |    25 |  | %rcx  |    26 |  | %rcx  |   126 |  | %rcx  |   126 |
| %rdx  | #1024 |  | %rdx  | #1024 |  | %rdx  | #1024 |  | %rdx  | #1024 |
| %r8   | #2048 |  | %r8   | #2048 |  | %r8   | #2048 |  | %r8   | #2048 |
| %r9   |     0 |  | %r9   |     0 |  | %r9   |     0 |  | %r9   |     2 |
|-------+-------|  |-------+-------|  |-------+-------|  |-------+-------|
| MEM   |       |  | MEM   |       |  | MEM   |       |  | MEM   |       |
| #1024 |   100 |  | #1024 |   100 |  | #1024 |   123 |  | #1024 |   123 |
| ...   |   ... |  | ...   |   ... |  | ...   |   ... |  | ...   |   ... |
| #2048 |   200 |  | #2048 |   200 |  | #2048 |   200 |  | #2048 |   201 |
| #2052 |   300 |  | #2052 |   300 |  | #2052 |   300 |  | #2052 |   335 |
| #2056 |   400 |  | #2056 |   400 |  | #2056 |   400 |  | #2056 |   400 |
|-------+-------|  |-------+-------|  |-------+-------|  |-------+-------|

addq $1,%rcx       addq (%rdx),%rcx   addl $1,(%r8,%r9,4)
addq %rbx,%rax     addq %rbx,(%rdx)   addl $1,%r9d
                   addq $3,(%rdx)     addl %eax,(%r8,%r9,4)
                                      addl $1,%r9d
                                      addl (%r8,%r9,4),%eax
```

The Other ALU Instructions

- Most ALU instructions follow the same patter as `addX`: two operands, second gets changed.
- Some one operand instructions as well.

Instruction	Name	Effect	Notes
`addX B, A`	Add	`A = A + B`	Two Operand Instructions
`subX B, A`	Subtract	`A = A - B`	
`imulX B, A`	Multiply	`A = A * B`	*Has a limited 3-arg variant*
`andX B, A`	And	`A = A & B`	
`orX B, A`	Or	`A = A | B`	
`xorX B, A`	Xor	`A = A ^ B`	
`salX B, A`	Shift Left	`A = A << B`	`B` is constant or `%cl` reg
`shlX B, A`		`A = A << B`	
`sarX B, A`	Shift Right	`A = A >> B`	Arithmetic: Sign carry
`shrX B, A`		`A = A >> B`	Logical: Zero carry
`incX A`	Increment	`A = A + 1`	One Operand Instructions
`decX A`	Decrement	`A = A - 1`	
`negX A`	Negate	`A = -A`	
`notX A`	Complement	`A = ~A`	

leaX: Load Effective Address

- Memory addresses must often be loaded into registers
- Often done with a leaX, usually leaq in 64-bit platforms
- Sort of like "address-of" op & in C but a bit more general

```
INITIAL
|-------+-------|
| REG   |   VAL |
| rax   |     0 |
| rcx   |     2 |
| rdx   | #1024 |
| rsi   | #2048 |
|-------+-------|
| MEM   |       |
| #1024 |    15 |
| #1032 |    25 |
| ...   |       |
| #2048 |   200 |
| #2052 |   300 |
| #2056 |   400 |
|-------+-------|
```

```
## leaX_examples.s:
movq  8(%rdx),%rax          # rax = *(rdx+1)     = 25
leaq  8(%rdx),%rax          # rax = rdx+1        = #1032
movl  (%rsi,%rcx,4),%eax # rax = rsi[rcx]     = 400
leaq  (%rsi,%rcx,4),%rax # rax = &(rsi[rcx]) = #2056
```

Compiler sometimes uses leaX for multiplication as it is usually faster than imulX but less readable.

```
# Odd Collatz update n = 3*n+1
#READABLE with imulX    #OPTIMIZED with leaX:
imul $3,%eax            leal 1(%eax,%eax,2),%eax
addl $1,%eax
# eax = eax*3 + 1       # eax = eax + 2*eax + 1,
# 3-4 cycles            # 1 cycle
```

Division: It's a Pain (1/2)

- `idivX` operation has some special rules
- Dividend must be in the `rax` / `eax` / `ax` register
- Sign extend to `rdx` / `edx` / `dx` register with "preparation" instructions like `cltd`
- `idivX` takes one **register** argument which is the divisor
- At completion
  - `rax` / `eax` / `ax` holds quotient (integer part)
  - `rdx` / `edx` / `dx` holds the remainder (leftover)

```
### division_example.s: start with uninitialized registers
## rax: 0x???????? = ? rdx: 0x???????? = ?
movl $5, %ecx            # will divide by 5
movl $19, %eax           # divide `int a=19;` by 5
cltd                     # prepare for 32-bit division
## combined 64-bit register %edx:%eax is
## rax: 0x0000000F = 15  rdx: 0x00000000 =  0
idivl %ecx               # 32-bit signed division
## 19 div 5 = 3 rem 4
## %eax == 3, quotient  %edx == 4, remainder
```

Compiler avoids division whenever possible: compile `col_unsigned.c` and `col_signed.c` to see some tricks involving shifts.

Division: It's a Pain (2/2)

Division / preparation is irregular. See `division_examples.s` for more details which are summarized below.

```
movq $23, %rax          # divide `long a=23;` by 5
cqto                    # prepare for 64-bit division
idivq %rcx              # rax = 4, rdx = 3

movl $19, %eax          # divide `int a=19;` by 5
cltd                    # prepare for 32-bit division
idivl %ecx              # eax = 3, edx = 4

movw $37, %ax           # divide `short a=37;` by 5
cwtd                    # prepare for 16-bit division
idivw %cx               # ax = 7, dx = 2

## 8-Bit is a SPECIAL CASE
movb $42, %al           # divide `char a=42;` by 5
cbtw                    # prepare for 8-bit division
idivb %cl               # al = 8, ah = 2
## NOTE: quotient AND remainder in ax-family reg
```

Preparation instructions like `cltd` copy the sign bit from the AX-family register elsewhere: 0 for positive numbers, 1 for negative numbers, crucial for hardware division algorithm.